From Documentation to Knowledge Vessels: Agentic AI for Tacit Knowledge Preservation in Manufacturing

Michael Banf*, Johannes Kuhn, Philipp Uhlmann, Emily Rasenecker, Max Becker and Sebastian Fetz

Perelyn GmbH, Munich, Germany

Abstract

Manufacturing faces an unprecedented knowledge crisis: an estimated 70% of critical operational expertise is undocumented and resides in aging workers' heads. Current AI-based knowledge capture treats this as a *documentation problem*—extracting expertise into static SOPs, knowledge graphs, or video tutorials. In this position paper, we argue for a fundamental reframing: tacit knowledge preservation should be understood as an *agent memory problem*. Drawing on a landscape analysis of 17+ commercial platforms and the agentic memory literature, we identify a critical *episodic-temporal gap*: no existing system captures the situated, evolving, episode-grounded nature of expert knowledge. We sketch a three-stage architecture—Capture–Organize–Transfer—in which graph-based memory structures (episode graphs, knowledge graphs, procedure trees) serve as the unifying representation at every layer, enabling agentic AI to incrementally absorb, structure, and deliver expertise as persistent *knowledge vessels*. Rather than presenting a finished implementation, we lay out this architecture as an invitation to the community: to discuss the design space, identify overlooked challenges, and collaboratively advance memory-grounded knowledge preservation in manufacturing.

Keywords

Tacit knowledge, Manufacturing, Agentic AI, Knowledge preservation, Agent memory, Multimodal capture

1. Introduction

The manufacturing sector confronts a structural knowledge crisis driven by workforce demographics. In the United States, 52.4% of the industrial products workforce is expected to retire or depart within five years [1], creating an estimated 2.8 million retirement-driven vacancies by 2033 [5]. The average age in small manufacturing shops approaches 55, with ownership often in their 60s [7]. Each retiring technician with 20+ years of experience takes approximately \$800K–\$1.2M in undocumented troubleshooting knowledge [6]. Fortune warned in January 2026 that there are an estimated 1–2 years left to capture decades of industrial knowledge [10]. An estimated 70% of critical manufacturing knowledge is never formalized [8], and knowledge transfer failures cost large manufacturers \$47 million per year [17]. When tribal knowledge is lost, repair times increase 3–4× and new-hire ramp-up extends from 3–4 months to 8–12 months [6].

A rapidly growing ecosystem of commercial platforms and academic prototypes has emerged to address this crisis. Yet virtually all treat tacit knowledge preservation as a *documentation problem*: conduct interviews, record SOPs, transcribe video, and store results as static artifacts. This framing discards the very qualities that make expert knowledge valuable, i.e. its situatedness, temporal evolution, and episode-grounded nature.

We propose a conceptual reframing: tacit knowledge preservation is not a documentation problem but an *agent memory problem*. Recent surveys reveal that all major agent harnesses lack episodic memory and temporal versioning [2], precisely the capabilities needed to capture how expert knowledge is situated, evolves over time, and is grounded in specific episodes. This reframing is not merely semantic:

Third Workshop on AI in Production, co-located with IJCAI 2026, August 11, 2026, Bremen, Germany
*Corresponding author.
michael.banf@perelyn.com (M. Banf)

Approach	Examples	Epis.	Temp.	Proc.	Consol.	Agent.
Current Approaches						
Process mining	Eschbach, Expl. ML	✗	✗	✗	✗	✗
Eye/motion tracking	Tobii/Denso, Fraunh. IPA	✗	✗	✗	✗	✗
Interview → KG	Squirro, Sugarwork	◆	✗	✗	✗	✗
Voice/Video → SOP	Dovient, Augmentir, OxMaint	✗	✗	✓	✗	✗
Connected worker	Poka, MaintainX, Tulip	✗	✗	◆	✗	✗
Document → semantic	Blockbrain, Glean, eGain	✗	✗	✗	✗	✗
Cognitive DT	COGNIMAN, Siemens	✗	◆	✗	✗	✗
Agentic diagnosis	Bosch Shopfloor Ag.	◆	✗	✗	✗	✓
Agent interviews	Zuin et al.	◆	✗	✗	✗	✓

✓ Supported ◆ Partial ✗ Absent

Figure 1: Feature matrix of knowledge capture approaches assessed against five memory capabilities derived from the CoALA taxonomy [20]. No surveyed system achieves more than two of five capabilities.

it shifts the design space from static capture-and-store pipelines to *living memory architectures* that continuously absorb, structure, and deliver expertise, even long after the original expert has retired.

We sketch a three-stage architecture (Capture–Organize–Transfer) and argue that *graph-based representations*—episode graphs, knowledge graphs, procedure trees—provide the natural unifying substrate at every layer. Rather than reporting a completed system, this paper lays out the architectural vision and invites the community to discuss its feasibility, identify blind spots, and collaborate on realizing memory-grounded knowledge preservation in manufacturing.

2. Current Landscape

We surveyed the current landscape of AI-based tacit knowledge preservation in manufacturing across three categories: commercial platforms, academic prototypes, and industry initiatives (Table 1).

At least 17 commercial products now target manufacturing knowledge capture, from dedicated startups—Dovient (multimodal capture, reducing diagnosis time from 25 min to 90 sec), Murray Mentor (voice-first in-ear guidance), Blockbrain (Stuttgart; Knowledge Bots for Bosch and Kärcher)—to mature connected-worker platforms (Augmentir, MaintainX, Poka with 2,300+ customers) and enterprise vendors (Siemens' Industrial Foundation Model trained on 150 PB, SAP's "Company Memory" context graph, IBM Maximo 9.1 with watsonx.ai). On the research side, Kernan Freire et al. deployed an LLM-based knowledge-sharing system in real factories, finding that workers perceived clear benefits but *still preferred learning from a human expert when available* [11]—a result we consider central to system design. Zuin et al. built an LLM agent that iteratively interviews employees to reconstruct organizational knowledge, achieving 94.9% full-knowledge recall in simulation [23]. Radhakrishnan demonstrated a conversational AI (Gemini 2.5 Pro) that converts tacit process knowledge into BPMN diagrams within a 12-minute session at SME-friendly API costs [19]. The EU Horizon project COGNIMAN develops cognitive digital twins integrating memory, perception, and reasoning [4]. Fraunhofer IESE has deployed dialogue-based AI assistants for knowledge capture in pump manufacturing, crane systems, and switch cabinet construction. The BMBF-funded KI_eeper project built a multimodal system for automatic expert knowledge capture in SMEs, finding that 80% of business-relevant knowledge remains undocumented [14]. Yet none of these projects implements episodic memory or temporal versioning. Notably, Horizon Europe's 2027 work programme now explicitly calls for systems that "capture and formalise the knowledge which is dispersed and not explicit" [12]—acknowledging that despite over a decade of EU funding, the problem remains unsolved.

Figure 1 formalizes the assessment across five memory capabilities derived from the CoALA taxonomy [20] and the bi-temporal model [26]. Despite the breadth of the ecosystem, *no* surveyed system achieves more than two of five capabilities: voice/video platforms capture procedures but discard context; agentic systems seek knowledge actively but store it as flat artifacts; cognitive digital twins track machine state evolution but not human expertise. Most critically, none implements episodic memory with temporal versioning or consolidation of episodes into generalized knowledge. This mirrors the *episodic-temporal paradox* in the broader agentic memory literature: 59% of recent research papers address episodic memory conceptually but only 4% implement temporal mechanisms beyond timestamps [2].

3. Framework: Capture–Organize–Transfer

We propose a three-stage framework (Figure 2) in which agentic AI orchestrates tacit knowledge preservation as a continuous, minimally invasive process rather than a one-time documentation effort. A key architectural observation is that *graph-based representations* emerge as the natural substrate at every stage: episodes are stored as richly connected event graphs linking actors, equipment, conditions, and outcomes; consolidated knowledge forms a semantic knowledge graph grounded in domain ontologies; and procedures are represented as decision trees with provenance edges back to their source episodes. This convergence on graphs is not coincidental—it reflects the fundamentally *relational* nature of manufacturing expertise, where understanding a failure requires traversing connections between equipment state, environmental conditions, material properties, and human decisions [24].

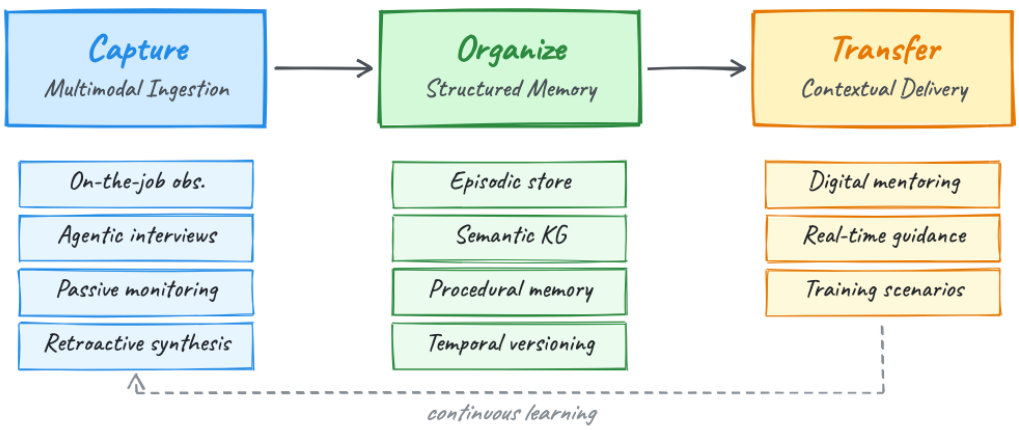

Figure 2: The Capture–Organize–Transfer framework. A feedback loop enables the agent to refine its knowledge continuously from new worker interactions.

3.1. Stage 1: Capture—Minimally Invasive Multimodal Ingestion

A core design principle is *minimal invasiveness*: experts are scarce, busy, and cannot be pulled from the production floor for lengthy knowledge-engineering sessions. The agentic system must capture knowledge *while experts work*, combining multiple modalities. We identify an *invasiveness spectrum* ranging from fully passive to lightly interactive, each targeting different knowledge layers:

Passive process mining (zero worker effort). Existing machine logs, shift records, and maintenance tickets contain latent expert knowledge. Eschbach's Shiftconnector mines years of accumulated shift logs to surface tacit patterns retroactively. Explainable ML pipelines can differentiate expert from novice behavior from process data alone with >90% accuracy, requiring no worker instrumentation [9]. This layer captures *what* experts do differently, but not *why*.

Ambient observation (no behavior change). Computer vision monitors expert actions during normal production; wearable eye trackers reveal visual attention strategies that experts cannot verbalize. Denso deployed Tobii eye-tracking glasses on inspectors, uncovering gaze patterns that explained exceptional quality and halving training time [21]. This layer captures embodied knowledge that resists articulation—a partial answer to Polanyi's paradox.

Agentic micro-interviews (brief, contextual). Rather than lengthy sessions, the agent conducts brief elicitation at natural pause points—shift changes, after anomaly resolution, or during routine checks. The Bosch Shopfloor Agent, deployed at Bosch Bamberg, uses a multi-agent system with voice interaction to diagnose root causes from historical data, saving €850K per plant annually [3]. Crucially, the agent *asks* rather than only *answers* [15], proactively filling knowledge gaps.

Retroactive synthesis. The system continuously ingests shift handoff notes, chat messages, and informal documentation—what Uchihira [22] calls "digital fragmented knowledge"—and links expert actions to contemporaneous sensor data, capturing the physical context that triggered intervention.

The key distinction from existing approaches is that the capture agent operates *autonomously and continuously* across all layers, identifying what it does not know and opportunistically filling gaps—rather than relying on scheduled documentation sessions.

3.2. Stage 2: Organize—Temporally Versioned Structured Memory

Following the cognitive memory taxonomy [20], captured knowledge is organized into three interconnected, temporally versioned stores (Table 3).

Episodic memory stores specific situated events with full context: who acted, what happened, when and where, what machine state triggered the intervention, and what the outcome was. Each episode is represented as a structured node carrying the tuple $\langle$*trigger, context, action, outcome, actor, timestamp*$\rangle$, linked to contemporaneous sensor snapshots and machine state vectors. This is the layer most critically absent from current systems: no surveyed platform stores the situated event "bearing on Press 7 failed after weekend shutdown in high humidity—senior technician diagnosed misalignment, not wear" as a retrievable, contextually grounded episode. Architecturally, episodic stores require graph-based representations where episodes connect to equipment nodes, personnel, and environmental conditions—flat vector stores cannot represent the relational structure of situated events [24].

Semantic memory consolidates abstracted knowledge into a manufacturing knowledge graph grounded in domain ontologies (e.g., ISA-95 for process hierarchies, ISO 14224 for failure classification). Nodes represent equipment, materials, failure modes, and causal relationships; edges encode dependencies ("polymer X requires pressure $\leq$3.8 bar") and constraints. Crucially, semantic nodes are *derived from* episodic evidence: the generalization "weekend shutdowns increase bearing misalignment risk on hydraulic presses" emerges only after multiple episodes are consolidated. This episodic-to-semantic consolidation mirrors how human experts build intuition from accumulated experience [25].

Procedural memory encodes *how* to perform tasks as step sequences, decision trees, and conditional heuristics—including experts' rules of thumb that deviate from official SOPs. Each procedure carries version metadata and links to the episodic evidence from which it was derived: "Step 3 was added after the incident on 2024-03-15 (Episode #247)." This provenance chain means procedures are not opaque prescriptions but *auditable, traceable* instructions whose rationale can be inspected—a requirement for safety-critical manufacturing and EU AI Act compliance.

Bi-temporal versioning. All three stores must be **temporally versioned** using a bi-temporal model [26] that distinguishes *event time* (when something happened on the shop floor) from *ingestion time* (when the system learned it). Every knowledge edge carries an explicit validity interval $[t_{\text{valid\_from}}, t_{\text{valid\_to}})$. When a material substitution changes process parameters, the old edge is closed and a new one opened—preserving the full history. This enables temporal queries that no current system supports: "What was the recommended pressure *before* the polymer switch in March 2024?" or "When did we first learn that humidity affects bearing alignment?"

Consolidation loop. A periodic consolidation process promotes recurring patterns from episodic to semantic memory and distills validated action sequences into procedural memory, while retaining the source episodes as provenance. This mirrors the hierarchical consolidation formalized in recent memory theory [27]: raw episodes are never discarded, but increasingly abstract representations are layered on top, enabling retrieval at the appropriate granularity—from a specific incident (episodic) to a general principle (semantic) to an executable procedure (procedural).

3.3. Stage 3: Transfer—Contextual Delivery to New Workers

The value of preserved knowledge is realized through effective transfer. The agentic system supports three delivery modalities: (1) **Digital mentoring**—the agent acts as a virtual experienced colleague, answering not just "how" but "why" by retrieving relevant episodes ("The last time someone saw this vibration on Press 7 after a humid weekend, it was bearing misalignment—here's what the senior technician did"); (2) **Real-time guidance**—in-ear or on-screen support calibrated to the worker's experience level, from procedural steps (novices) to contextual warnings (intermediates); and (3) **Scenario-based training**—past episodes recomposed into exercises that expose new workers to rare edge cases omitted from official materials. Crucially, the feedback loop (Figure 2) ensures the knowledge vessel *continues learning* from new workers' experiences, extending the knowledge base beyond what any single expert contributed.

4. Illustrative Walkthrough: Bearing Failure After Material Substitution

To ground the framework concretely, we trace a representative manufacturing scenario through all three stages. This walkthrough illustrates what no current system provides: the connection between a situated episode, its temporal context, the generalized knowledge derived from it, and the executable procedure grounded in that evidence.

Scenario. A hydraulic press (Press 7) uses polymer seals from Supplier A with a recommended operating pressure of 4.2 bar. In March 2024, procurement switches to Supplier B's polymer, which requires lower pressure. Over subsequent months, a senior technician discovers through trial and error that 3.8 bar is optimal, and that weekend shutdowns in humid conditions cause bearing misalignment (not wear, as the maintenance manual suggests). In September, a partial reversion to 4.0 bar proves necessary due to batch variability. The technician retires in December 2024.

Stage 1 (Capture) The knowledge vessel captures this expertise through multiple modalities: (i) *passive mining* detects the pressure parameter change in machine logs and correlates it with the supplier switch in procurement records; (ii) *ambient observation* via vibration sensors flags anomalous bearing behavior after weekend shutdowns; (iii) an *agentic micro-interview* at shift change—triggered by the anomaly—elicits the technician's diagnosis: "It's misalignment from thermal contraction, not wear. Check alignment before you check the bearing." None of these modalities alone captures the full picture; the agent synthesizes across all three.

Stage 2 (Organize) The captured knowledge populates all three memory stores: *Episodic*: three distinct event nodes record the initial failure (March), the humidity discovery (July), and the pressure reversion (September), each linked to sensor snapshots and the technician's explanations. *Semantic*: the knowledge graph gains edges $\texttt{PolymerB} \xrightarrow{\text{requires}} \leq\texttt{3.8bar}$ with validity [Mar 2024, Sep 2024) and $\texttt{PolymerB} \xrightarrow{\text{requires}} \texttt{4.0bar}$ from [Sep 2024, now), plus a causal edge $\texttt{humid\_weekend} \xrightarrow{\text{causes}} \texttt{bearing\_misalignment}$ on hydraulic presses. *Procedural*: the restart checklist gains a new step: "After weekend shutdown in >60% RH: verify bearing alignment before production start (source: Episodes #247, #251)."

Stage 3 (Transfer) Six months after the technician's retirement, a new operator encounters unusual vibration on Press 7 after a rainy weekend. The knowledge vessel retrieves the relevant episodes and provides contextual guidance: "Similar vibration was diagnosed as bearing misalignment (not wear) by [senior technician] in July 2024 under similar humidity conditions. Recommended action: check alignment per Step 3a of the restart checklist. Historical episodes: #247, #251." The operator follows the procedure, confirms misalignment, and resolves the issue in minutes rather than hours. A conventional SOP would have directed them to replace the bearing.

Memory type	Manufacturing example	Representation	Architectural basis
Episodic	Bearing failed on Press 7 after humid weekend; technician found misalignment, not wear	Event graph node with sensor snapshot, actor, outcome	Graph-based with temporal edges (Zep)
Semantic	Polymer X requires ≤3.8 bar; weekend shutdowns increase misalignment risk on hyd. presses	KG triples grounded in ISA-95 / ISO 14224 ontology	Ontology-grounded KG with validity intervals
Procedural	After weekend shutdown: check bearing alignment before restart (added after Episode #247)	Versioned step sequence with provenance links	Decision tree + episode references (MemP)

All stores temporally versioned with bi-temporal validity intervals

Figure 3: Architectural mapping: memory types, manufacturing knowledge, and realization. Each store is temporally versioned with bi-temporal validity intervals [26, 28].

5. Discussion: Challenges and Open Questions

The architecture outlined above is deliberately a *design proposal*, not a finished system. We identify four open challenges that we believe warrant community discussion.

Polanyi's observation that "we can know more than we can tell" [18] remains a fundamental constraint. AI can capture *expressible* tacit knowledge—heuristics, rules of thumb, decision rationales—but deeply embodied knowledge (the "feel" of a machine vibration) resists formalization. Ambient sensing (eye tracking, motion capture) can approximate the physical correlates of embodied skill but cannot fully replicate the experience. *Where exactly does the boundary lie for manufacturing knowledge, and how should a system communicate its own epistemic limits?*

Ide [13] formalizes a counterintuitive risk: AI systems that reduce the need for human mentoring may *weaken* the apprenticeship pipeline through which deep tacit knowledge is transmitted. Our framework positions the agent as a *complement* to human mentoring—consistent with the empirical finding that workers prefer human experts when available [11]. *But how do we design systems that strengthen rather than substitute human knowledge transfer?*

In safety-critical manufacturing, incorrect knowledge transfer can cause physical harm. Memory poisoning has been identified as a top agentic risk, with demonstrated injection rates exceeding 95% [16]. The EU AI Act (August 2026) classifies AI in safety-critical manufacturing as high-risk, mandating transparency, human oversight, and tamper-evident audit trails. *What validation mechanisms are sufficient for graph-based memory architectures in high-stakes environments? How should provenance and temporal versioning interact with regulatory audit requirements?*

The most acute knowledge concentration risk is in SMEs, where a single retirement can eliminate critical capabilities. Lightweight, voice-first interfaces—Murray Mentor's in-ear guidance and Radhakrishnan's 12-minute BPMN capture [19]—demonstrate that effective capture need not require heavy IT infrastructure. *Can the graph-based memory architecture we propose be realized in resource-constrained settings, or does it inherently require enterprise-scale infrastructure?*

6. Conclusion

The manufacturing sector's tacit knowledge crisis is urgent, costly, and accelerating. While 17+ commercial platforms address knowledge capture, the entire field operates within a *documentation paradigm* that discards the episodic, temporal, and situated qualities that make expert knowledge most valuable.

In this position paper, we have argued for reframing tacit knowledge preservation as an agent memory problem and sketched the Capture–Organize–Transfer architecture, in which graph-based memory structures—episode graphs, knowledge graphs, procedure trees—provide the unifying representation at every layer. The gap we identify is actionable: agentic AI infrastructure and manufacturing knowledge management both exist in mature form, but have not been connected through an episodic-temporal memory architecture.

We close with three questions we hope will spark discussion: *(i)* What is the minimal viable memory architecture for manufacturing knowledge vessels—can episodic-temporal grounding be achieved without full graph infrastructure? *(ii)* How should consolidation from episodic to semantic memory be validated in safety-critical domains where incorrect generalizations carry physical risk? *(iii)* Can the reframing proposed here generalize beyond manufacturing to other domains facing tacit knowledge loss (e.g., healthcare, skilled trades, engineering)? We invite the community to engage with these questions and to collaboratively shape the research agenda for memory-grounded knowledge preservation.

References

[1] APQC: Knowledge exodus from the silver tsunami in the industrial products sector: It is AI or bust. ATD Blog (2026)

[2] Banf, M., et al.: Memory-aware software engineering agents: Architectures, mechanisms, and open gaps. AI4SE Workshop @ KI 2026 (2026)

[3] Bosch: Agentic AI in manufacturing production—Shopfloor Agent. Bosch Stories (2025)

[4] Belbachir, A., et al.: COGNIMAN digital twin architecture for flexible manufacturing. J. Intell. Manuf. (2025)

[5] Deloitte, NAM: Creating pathways for tomorrow's workforce today. Manufacturing Institute Report (2024)

[6] Dovient: The real cost of tribal knowledge loss in manufacturing. White Paper (2025)

[7] Edwards, K.M., et al.: Agentic AI in engineering and manufacturing. arXiv:2604.09633 (2026)

[8] eGain: Capturing tacit knowledge from the great retirement cohort using GenAI. Report (2025)

[9] de Arriba-Pérez, F., et al.: Automatic generation of insights from workers' actions in industrial workflows with explainable Machine Learning. arXiv:2406.12732 (2024)

[10] Sharma, K.: Let's train workers on industrial AI, not replace them. Fortune (Jan 2026)

[11] Kernan Freire, S., et al.: Knowledge sharing in manufacturing using LLM-powered tools. Frontiers in AI **7** (2024)

[12] European Commission: Horizon Europe Work Programme 2026–2027, Topic HORIZON-CL4-2027-02-DIGITAL-EMERGING-52: New approaches for Human/AI collaboration (2026)

[13] Ide, E.: Automation, AI, and the intergenerational transmission of knowledge. arXiv:2507.16078 (2026)

[14] Ottersböck, N., et al. (eds.): Wissensmanagement mit Künstlicher Intelligenz. Springer (2025)

[15] Najjar, R.: From AI-based KM assistant to tacit knowledge agent: Leveraging small language models for tacit knowledge extraction. Medium (2025)

[16] OWASP: Agentic security initiative—ASI06: Memory poisoning. OWASP Foundation (2026)

[17] Panopto: Workplace knowledge and productivity report (2018)

[18] Polanyi, M.: The Tacit Dimension. Univ. of Chicago Press (1966)

[19] Radhakrishnan, U.: Documenting SME processes with conversational AI. arXiv:2512.05122 (2025)

[20] Sumers, T.R., et al.: Cognitive architectures for language agents. Trans. Mach. Learn. Res. (2024)

[21] Tobii: Unlocking hidden skills—eye tracking for workforce training. Tobii Blog (2025)

[22] Uchihira, N.: Tacit knowledge management with generative AI: The GenAI SECI model. arXiv:2603.21866 (2026)

[23] Zuin, G., et al.: Leveraging LLMs for tacit knowledge discovery. IJCNN (2025)

[24] Yang, C., et al.: Graph-based agent memory: A survey. arXiv:2602.05665 (2026)

[25] Wu, Z., et al.: GAM: Hierarchical graph memory for LLM-based agents. In: ICLR 2026

[26] Rasmussen, P., et al.: Zep: Temporal knowledge graph for agent memory. arXiv:2501.13956 (2025)

[27] TiMem: Temporal-hierarchical memory consolidation for language agents. arXiv:2601.02845 (2026)

[28] Fang, R., et al.: Mem^p: Exploring agent procedural memory. arXiv:2508.06433 (2025)